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Computer Science & Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 14/09/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	Andhra Pradesh Board of Intermediate Education	Sri Chaitanya	2021	98.30%
Matriculation	Board Of Secondary Education Andhra Pradesh	Sri Chaitanya	2019	10

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 377** in JEE Advanced 2021 amongst the 1,50,000 candidates ('21)
- Secured **All India Rank 543** in JEE Mains 2021 amongst 1.2 Million candidates ('21)
- Recipient of **National Talent Search Examination (NTSE)** Scholarship by NCERT ('19)
- Secured **State Rank 17** in AP-EAMCET conducted by Andhra Pradesh State Council ('21)
- Secured **State Rank 126** in TS-EAMCET conducted by Telangana State Council ('21)

INTERNSHIP EXPERIENCE

Qualcomm | Software Development Intern

(Summer '24)

- Worked on **heterogeneous computing**, utilizing **RPC** to efficiently offload computational tasks to remote processors
- Debugged** various issues in the **open-source** user-space code of the FastRPC framework, improving system stability
- Developed **kernel patches** for FastRPC driver, soon to be **upstreamed**, enhancing the functionality of the driver

KEY PROJECTS

Scpl C-like Compiler

(Spring '24)

Guide: Prof. Uday Khedker | Course Project: Implementation of Programming Languages

IIT Bombay

- Developed a C subset **compiler** supporting **functions, scopes, control flow**, and optimizations with error handling
- Utilized **Lex** for tokenizing and **Yacc** for syntactic parsing to construct **ASTs** and generate **Three Address Codes**
- Designed **RTL** and **TAC** classes using the Object Oriented Programming paradigm, carefully following class hierarchies

Improved IPCP Prefetcher

(Spring '23)

Guide: Prof. Biswabandan Panda | Course Project : DLD and Computer Architecture

IIT Bombay

- Obtained an average **5% speed-up in IPC** on various traces by enhancing the IPCP prefetcher using Champsim
- Improved IP address classification by **increasing confidence bit allocation**, reducing misclassification errors
- Created a **hybrid order model** adaptable to changing confidence levels of IP classes for precise classification

Railway Planner

(Autumn '22)

Guide: Prof. Supratik Chakraborty | Course Project: DSA Lab

IIT Bombay

- Implemented various data structures and algorithms in C++, including **Linked Lists, Dictionaries, BSTs, Tries**
- Leveraged graph data structure to find the **shortest route** between two stations in the multi-rail network system
- Developed an intuitive interface for **rating and filtering system** for the reviews using **Heap and Priority Queues**

xv6 Optimization

(Autumn '23)

Guide: Prof. Purushottam Kulkarni | Course Project : Operating Systems

IIT Bombay

- Developed advanced **system calls**, and process **scheduler** algorithms to the xv6 code for improved performance
- Implemented **Lazy page allocation** with page fault handling functionalities to xv6 memory management system
- Implemented **threading, robust synchronization mechanisms**, and filesystem operations for concurrency in xv6

Cinema A to Z

(Autumn '22)

Guide: Prof. Kavi Arya | Course Project: Software Systems Lab

IIT Bombay

- Developed a comprehensive movie and TV show database by **web scraping** IMDb and Rotten Tomatoes for key data
- Designed, executed, and successfully deployed an **SQL database** to efficiently store and manage the scraped data
- Built a website in PHP for users to create accounts, log in, and **personalize their experience** by managing watchlists

Image Captioning

(Autumn '23)

Guide: Prof. Preethi Jyothi | Course Project: Artificial Intelligence and Machine Learning

IIT Bombay

- Implemented a research paper to build and train a deep model that **generates captions for images** using **PyTorch**.
- Implemented **visual attention**, enabling selective focus on key image areas and generating accurate captions
- Evaluated model performance using **BLEU** and **METEOR** scores to thoroughly assess caption quality and accuracy

OTHER PROJECTS

Cryptonite

(Spring '24)

Guide: Prof. Vinay Ribeiro | Course Project: Blockchains, Cryptocurrencies, and Smart Contracts

IIT Bombay

- Developed a discrete-time event simulator modeling a **blockchain cryptocurrency** network with honest miners
- Simulated message latency, **proof of work**, transaction generation, **verification**, and blockchain forks efficiently
- Incorporated node hyper-parameters in the network, evaluating the results of **selfish mining attacks** comprehensively

FaaS Application

(Spring '24)

Guide: Prof. Purushottam Kulkarni | Course Project : Topics in Virtualization and Cloud Computing

IIT Bombay

- Built a FaaS framework with **Kubernetes**, allowing clients to register and expose functions through a **Flask app**
- Allowed clients to seamlessly register custom functions via **Docker image or Python code** files, enhancing flexibility

Sliding Puzzle

(Spring '23)

Guide: Prof. Ashutosh Gupta | Course Project: Logic for Computer Science

IIT Bombay

- Developed SAT-based puzzle solver by implementing rules and constraints and finding **optimal moves** within limits
- Implemented efficient Python algorithm using **Z3 solver** and added **clauses for solving**, showcasing logical reasoning

Tic-Tac-Toe

(Autumn '22)

Guide: Prof. Kavi Arya | Course Project: Software Systems Lab

IIT Bombay

- Developed a two-player Tic-Tac-Toe game using a **Peer-to-Peer** networking model, using the Socket and Server
- Implemented message exchange mechanisms between players using ports, **enabling smooth real-time gameplay**

Credit card fraud Detection

(Spring '23)

Guide: Joshi Meet Anilkumar | Project: WIDS

Analytics Club, IIT Bombay

- **Analysed** the relations between the parameters and **visualized** the data using **pandas** and **matplotlib** effectively
- **Cleaned and balanced** the skewed data by using **SMOTE** and found an efficient model to fit the data effectively

Lattice based Cryptography

(Spring '23)

Guide: Prof. Manoj Prabhakaran | Course Project: Cryptography and Network Security

IIT Bombay

- Investigated mathematical **foundations of Lattices** and their relevance to cryptographic tools, including PQC
- Investigated lattice-based cryptosystems like **GGH** & **NTRU**, focusing on their foundations and resistance to attacks

Group Theory

(Spring '23)

Guide: Arpon Basu | Project : Summer of Science

Institute Technical Council, IIT Bombay

- Investigated the **mathematical principles** of groups, discovering fascinating insights into their **structure**, properties
- Explored **practical applications** of group theory in cryptography, physics, computer science, and network analysis

TECHNICAL SKILLS

Programming:	C++, C, Python, Assembly, Prolog, Haskell, Bash, Awk, Sed
Web Development:	HTML, CSS, Bootstrap, PHP, JavaScript
Software:	Git, L ^A T _E X, Doxygen, Sphinx, Jupyter, Docker, Wireshark
Packages:	NumPy, Matplotlib, Pandas, scikit-learn, Pytorch, Spacy

COURSES UNDERTAKEN

Computer Science:	Data Structures and Algorithms, Discrete Structures, Data Analysis and Interpretation, Software Systems Lab, Design and Analysis of Algorithms, Digital Logic Design, Computer Networks, Logic for Computer Science, Cryptography and Network Security, Abstractions and Paradigms in Programming, Computer Programming and Utilization, Artificial Intelligence and Machine Learning, Operating Systems, Speech and Natural Language Processing and the Web, Implementation of Programming Languages, Database and Information Systems, Topics in Virtualization and Cloud Computing
Mathematics:	Calculus, Linear Algebra, Differential Equations

EXTRACURRICULAR ACTIVITIES

- Successfully completed a course under the **National Sports Organization(NSO)** ('21)
- Actively participated as a volunteer in a **blood donation camp** organized by Abhuday, IIT Bombay. (Autumn '22)
- Secured 3rd position in the **Call of Duty** Tournament in CSE department organised by CSEA (Autumn '22)
- Participated in the **EnB Buzz** competition organized by **E-Cell, IIT Bombay** (Autumn '21)
- Participated in the **RC plane** competition 2022 powered by ideaForge, organised by Aeromodelling club (Autumn '22)
- Passionate about playing **Pool** and engaging in **Call of Duty**, which provide both strategic challenge and relaxation